

CHAPTER 5

elementary stochastic calculus



The aim of this Chapter...

... is to develop the theory behind the manipulation of random quantities, in particular stochastic differential equations like the one at the end of the previous chapter. This is very important for a thorough understanding of quantitative finance.

In this Chapter...

- all the stochastic calculus you need to know, and no more
- the meaning of Markov and martingale
- Brownian motion
- stochastic integration
- stochastic differential equations
- Itô's lemma in one and more dimensions

5.1 INTRODUCTION

Stochastic calculus is very important in the mathematical modeling of financial processes. This is because of the underlying random nature of financial markets. Because stochastic calculus is such an important tool I want to ensure that it can be used by everyone. To that end, I am going to try to make this chapter as accessible and intuitive as possible. By the end, I hope that the reader will know what various technical terms mean (and rarely are they very complicated), but, more importantly, will also know how to use the techniques with the minimum of fuss.

Most academic articles in finance have a 'pure' mathematical theme. The mathematical rigor in these works is occasionally justified, but more often than not it only succeeds in obscuring the content. When a subject is young, as is mathematical finance (*youngish*), there is a tendency for technical rigor to feature very prominently in research. This is due to lack of confidence in the methods and results. As the subject ages, researchers will become more cavalier in their attitudes and we will see much more rapid progress.



Time Out...

Aaaaarghhhhh!

If you don't feel comfortable with messy algebra, just skip to the end of the chapter where I outline the intuition behind stochastic calculus and give you a few rules of thumb to help you use it in practice. Actually, most of this chapter is just groundwork for the important mathematical 'tool' of Itô's lemma.

5.2 A MOTIVATING EXAMPLE

Toss a coin. Every time you throw a head I give you \$1, every time you throw a tail you give me \$1. Figure 5.1 shows how much money you have after six tosses. In this experiment the sequence was THHTHT, and we finished even.

If I use R_i to mean the random amount, either \$1 or -\$1, you make on the i th toss then we have

$$E[R_i] = 0, \quad E[R_i^2] = 1 \quad \text{and} \quad E[R_i R_j] = 0.$$

In this example it doesn't matter whether or not these expectations are conditional on the past. In other words, if I threw five heads in a row it does not affect the outcome of the sixth toss. To the gamblers out there, this property is also shared by a fair die, a balanced roulette wheel, but not by the deck of cards in blackjack. In blackjack the same deck is used for game after game, the odds during one game depend on what cards were dealt out from the same deck in previous games. That is why you can in the long run beat the house at blackjack but not roulette.

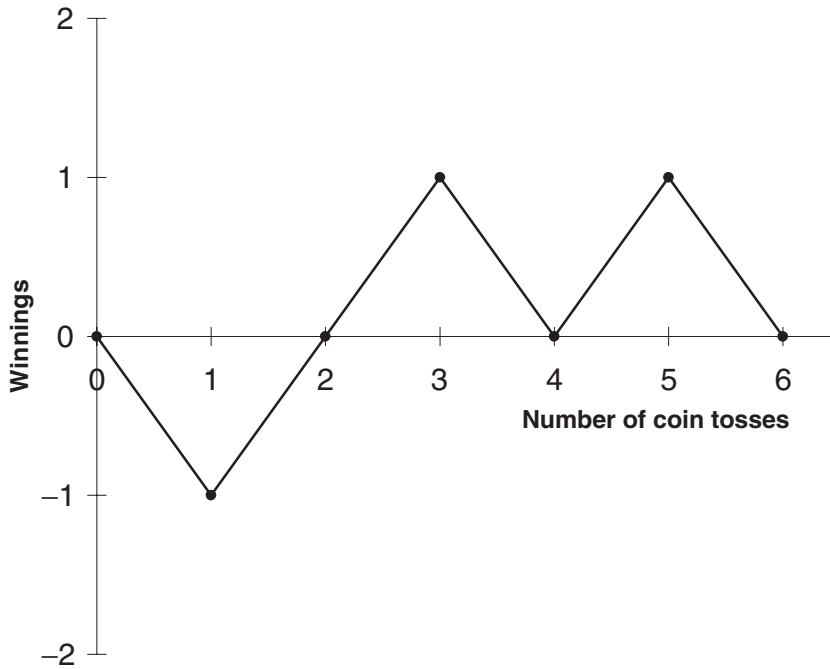


Figure 5.1 The outcome of a coin-tossing experiment.

Introduce S_i to mean the total amount of money you have won up to and including the i th toss so that

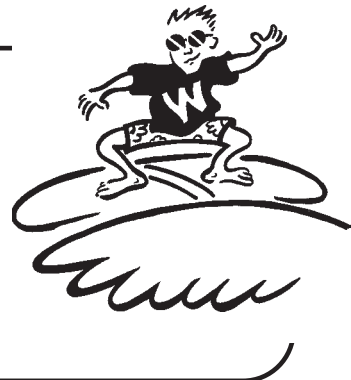
$$S_i = \sum_{j=1}^i R_j.$$

Later on it will be useful if we have $S_0 = 0$, i.e. you start with no money.

Time Out...

Just like coin tossing or the binomial tree

Very similar, but here we have something like an arithmetic random walk rather than geometric. . . we are adding or subtracting a quantity rather than multiplying.



If we now calculate expectations of S_i it *does* matter what information we have. If we calculate expectations of future events before the experiment has even begun then

$$E[S_i] = 0 \quad \text{and} \quad E[S_i^2] = E[R_1^2 + 2R_1R_2 + \dots] = i.$$

On the other hand, suppose there have been five tosses already, can I use this information and what can we say about expectations for the sixth toss? This is the **conditional expectation**. The expectation of S_6 conditional upon the previous five tosses gives

$$E[S_6|R_1, \dots, R_5] = S_5.$$



5.3 THE MARKOV PROPERTY

This result is special, the expected value of the random variable S_i conditional upon all of the past events *only depends on the previous value* S_{i-1} . This is the **Markov property**. We say that the random walk has no memory beyond where it is now. Note that it doesn't have to be the case that the expected value of the random variable S_i is the same as the previous value.

This can be generalized to say that given information about S_j for some values of $1 \leq j < i$ then the only information that is of use to us in estimating S_i is the value of S_j for the largest j for which we have information.

Almost all of the financial models that I will show you have the Markov property. This is of fundamental importance in modeling in finance. I will also show you examples where the system has a small amount of memory, meaning that one or two other pieces of information are important. And I will also give a couple of examples where *all* of the random walk path contains relevant information.

5.4 THE MARTINGALE PROPERTY

The coin-tossing experiment possesses another property that can be important in finance. You know how much money you have won after the fifth toss. Your expected winnings after the sixth toss, and indeed after any number of tosses if we keep playing, is just the amount you already hold. That is, the conditional expectation of your winnings at any time in the future is just the amount you already hold:

$$E[S_i|S_j, j < i] = S_j.$$

This is called the **martingale property**.

5.5 QUADRATIC VARIATION

I am now going to define the **quadratic variation** of the random walk. This is defined by

$$\sum_{j=1}^i (S_j - S_{j-1})^2.$$

Because you either win or lose an amount \$1 after each toss, $|S_j - S_{j-1}| = 1$. Thus the quadratic variation is always i :

$$\sum_{j=1}^i (S_j - S_{j-1})^2 = i.$$

I want to use the coin-tossing experiment for one more demonstration. And that will lead us to a continuous-time random walk.

5.6 BROWNIAN MOTION

I am going to change the rules of my coin-tossing experiment. First of all I am going to restrict the time allowed for the six tosses to a period t , so each toss will take a time $t/6$. Second, the size of the bet will not be \$1 but $\sqrt{t/6}$.

This new experiment clearly still possesses both the Markov and martingale properties, and its quadratic variation measured over the whole experiment is

$$\sum_{j=1}^6 (S_j - S_{j-1})^2 = 6 \times \left(\sqrt{\frac{t}{6}} \right)^2 = t.$$

I have set up my experiment so that the quadratic variation is just the time taken for the experiment.

I will change the rules again, to speed up the game. We will have n tosses in the allowed time t , with an amount $\sqrt{t/n}$ riding on each throw. Again, the Markov and martingale properties are retained and the quadratic variation is still

$$\sum_{j=1}^n (S_j - S_{j-1})^2 = n \times \left(\sqrt{\frac{t}{n}} \right)^2 = t.$$

I am now going to make n larger and larger. All I am doing with my rule changes is to speed up the game, decreasing the time between tosses, with a smaller amount for each bet. But I have chosen my new scalings very carefully, the time step is decreasing like n^{-1} but the bet size only decreases by $n^{-1/2}$.

In Figure 5.2 I show a series of experiments, each lasting for a time 1, with increasing number of tosses per experiment.

As I go to the limit $n = \infty$, the resulting random walk stays finite. It has an expectation, conditional on a starting value of zero, of

$$E[S(t)] = 0$$

and a variance

$$E[S(t)^2] = t.$$

I use $S(t)$ to denote the amount you have won or the value of the random variable after a time t . The limiting process for this random walk as the time steps go to zero is called **Brownian motion**, and I will denote it by $X(t)$.

The important properties of Brownian motion are as follows.

- **Finiteness:** Any other scaling of the bet size or ‘increments’ with time step would have resulted in either a random walk going to infinity in a finite time, or a limit in which there was no motion at all. It is important that the increment scales with the square root of the time step.
- **Continuity:** The paths are continuous, there are no discontinuities. Brownian motion is the continuous-time limit of our discrete time random walk.

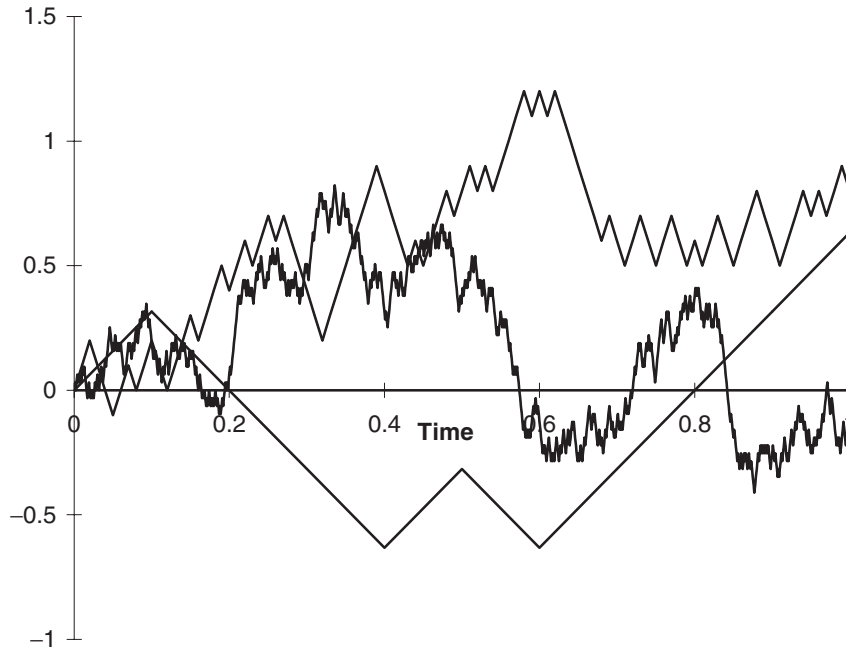


Figure 5.2 A series of coin-tossing experiments, the limit of which is Brownian motion.

- **Markov:** The conditional distribution of $X(t)$ given information up until $\tau < t$ depends only on $X(\tau)$.
- **Martingale:** Given information up until $\tau < t$ the conditional expectation of $X(t)$ is $X(\tau)$.
- **Quadratic variation:** If we divide up the time 0 to t in a partition with $n + 1$ partition points $t_i = it/n$ then

$$\sum_{j=1}^n (X(t_j) - X(t_{j-1}))^2 \rightarrow t. \quad (\text{Technically 'almost surely.'})$$

- **Normality:** Over finite time increments t_{i-1} to t_i , $X(t_i) - X(t_{i-1})$ is Normally distributed with mean zero and variance $t_i - t_{i-1}$.

Having built up the idea and properties of Brownian motion from a series of experiments, we can discard the experiments, to leave the Brownian motion that is defined by its properties. These properties will be very important for our financial models.

5.7 STOCHASTIC INTEGRATION

I am going to define a **stochastic integral** by

$$W(t) = \int_0^t f(\tau) dX(\tau) = \lim_{n \rightarrow \infty} \sum_{j=1}^n f(t_{j-1}) (X(t_j) - X(t_{j-1}))$$

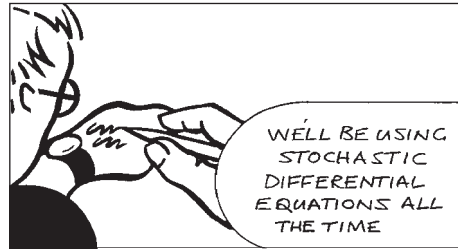
with

$$t_j = \frac{jt}{n}.$$

Before I manipulate this in any way or discuss its properties, I want to stress that the function $f(t)$ which I am integrating is evaluated in the summation at the *left-hand point* t_{j-1} . It will be crucially important that each function evaluation does not know about the random increment that multiplies it, i.e. the integration is **non anticipatory**. In financial terms, we will see that we take some action such as choosing a portfolio and only then does the stock price move. This choice of integration is natural in finance, ensuring that we use no information about the future in our current actions.

5.8 STOCHASTIC DIFFERENTIAL EQUATIONS

Stochastic integrals are important for any theory of stochastic calculus since they can be meaningfully defined. (And in the next section I show how the definition leads to some important properties.) However, it is very common to use a shorthand notation for expressions such as



$$W(t) = \int_0^t f(\tau) dX(\tau). \quad (5.1)$$

That shorthand comes from ‘differentiating’ (5.1) and is

$$dW = f(t) dX. \quad (5.2)$$

Think of dX as being an increment in X , i.e. a Normal random variable with mean zero and standard deviation $dt^{1/2}$.

Equations (5.1) and (5.2) are meant to be equivalent. One of the reasons for this shorthand is that the equation (5.2) looks a lot like an ordinary differential equation. We *do not* go the further step of dividing by dt to make it look exactly like an ordinary differential equation because then we would have the difficult task of defining $\frac{dX}{dt}$.

Pursuing this idea further, imagine what might be meant by

$$dW = g(t) dt + f(t) dX. \quad (5.3)$$

This is simply shorthand for

$$W(t) = \int_0^t g(\tau) d\tau + \int_0^t f(\tau) dX(\tau).$$

Equations like (5.3) are called **stochastic differential equations**. Their precise meaning comes, however, from the technically more accurate equivalent stochastic integral. In this book I will use the shorthand versions almost everywhere, no confusion should arise.

5.9 THE MEAN SQUARE LIMIT

I am going to describe the technical term **mean square limit**. This is useful in the precise definition of stochastic integration. I will explain the idea by way of the simplest example.

Examine the quantity

$$E \left[\left(\sum_{j=1}^n (X(t_j) - X(t_{j-1}))^2 - t \right)^2 \right] \quad (5.4)$$

where

$$t_j = \frac{jt}{n}.$$

This can be expanded as

$$E \left[\sum_{j=1}^n (X(t_j) - X(t_{j-1}))^4 + 2 \sum_{i=1}^n \sum_{j<i} (X(t_i) - X(t_{i-1}))^2 (X(t_j) - X(t_{j-1}))^2 - 2t \sum_{j=1}^n (X(t_j) - X(t_{j-1}))^2 + t^2 \right].$$

Since $X(t_j) - X(t_{j-1})$ is Normally distributed with mean zero and variance t/n we have

$$E \left[(X(t_j) - X(t_{j-1}))^2 \right] = \frac{t}{n}$$

and

$$E \left[(X(t_j) - X(t_{j-1}))^4 \right] = \frac{3t^2}{n^2}.$$

Thus (5.4) becomes

$$n \frac{3t^2}{n^2} + n(n-1) \frac{t^2}{n^2} - 2tn \frac{t}{n} + t^2 = O\left(\frac{1}{n}\right).$$

As $n \rightarrow \infty$ this tends to zero. We therefore say that

$$\sum_{j=1}^n (X(t_j) - X(t_{j-1}))^2 = t$$

in the ‘mean square limit.’ This is often written, for obvious reasons, as

$$\int_0^t (dX)^2 = t.$$

I am not going to use this result, nor will I use the mean square limit technique. However, when I talk about ‘equality’ in the following ‘proof’ I mean equality in the mean square sense.

5.10 FUNCTIONS OF STOCHASTIC VARIABLES AND ITÔ’S LEMMA

I am now going to introduce the idea of a function of a stochastic variable. In Figure 5.3 is shown a realization of a Brownian motion $X(t)$ and the function $F(X) = X^2$.

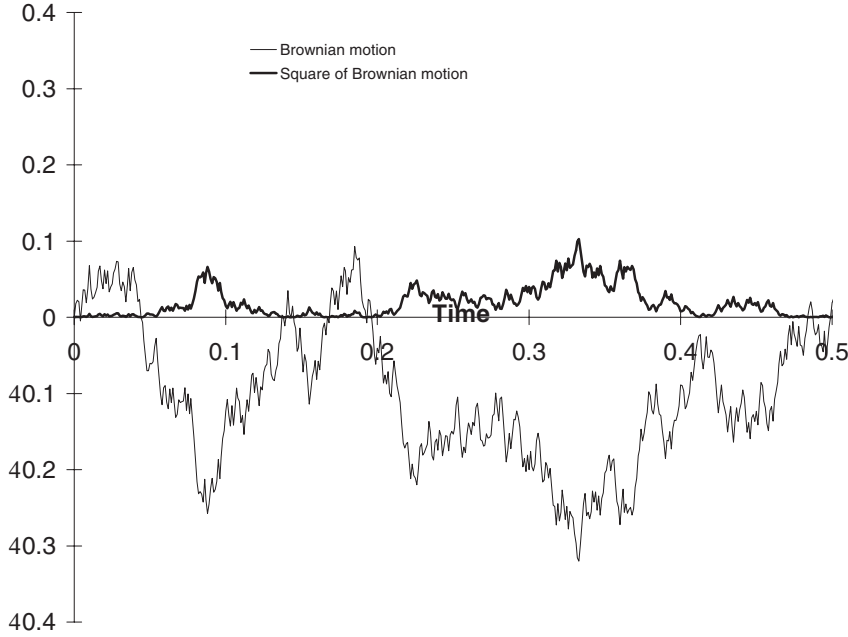


Figure 5.3 A realization of a Brownian motion and its square.

If $F = X^2$ is it true that $dF = 2X dX$? No. The ordinary rules of calculus do not generally hold in a stochastic environment. Then what are the rules of calculus?

I am going to ‘derive’ the most important rule of stochastic calculus, **Itô’s lemma**. My derivation is more heuristic than rigorous, but at least it is transparent. I will do this for an arbitrary function $F(X)$.

In this derivation I will need to introduce various timescales. The first timescale is very, very small. I will denote it by

$$\frac{\delta t}{n} = h.$$

This timescale is so small that the function $F(X(t+h))$ can be approximated by a Taylor series:

$$F(X(t+h)) - F(X(t)) = (X(t+h) - X(t)) \frac{dF}{dX}(X(t)) + \frac{1}{2}(X(t+h) - X(t))^2 \frac{d^2F}{dX^2}(X(t)) + \dots$$

From this it follows that

$$\begin{aligned} & (F(X(t+h)) - F(X(t))) + (F(X(t+2h)) - F(X(t+h))) + \dots + (F(X(t+nh)) \\ & - F(X(t+(n-1)h))) \\ &= \sum_{j=1}^n (X(t+jh) - X(t+(j-1)h)) \frac{dF}{dX}(X(t+(j-1)h)) \\ &+ \frac{1}{2} \frac{d^2F}{dX^2}(X(t)) \sum_{j=1}^n (X(t+jh) - X(t+(j-1)h))^2 + \dots \end{aligned}$$

In this I have used the approximation

$$\frac{d^2 F}{dX^2}(X(t + (j-1)h)) = \frac{d^2 F}{dX^2}(X(t)).$$

This is consistent with the order of accuracy I require.

The first line in this becomes simply

$$F(X(t + nh)) - F(X(t)) = F(X(t + \delta t)) - F(X(t)).$$

The second is just the definition of

$$\int_t^{t+\delta t} \frac{dF}{dX} dX$$

and the last is

$$\frac{1}{2} \frac{d^2 F}{dX^2}(X(t)) \delta t,$$

in the *mean square sense*. Thus we have

$$F(X(t + \delta t)) - F(X(t)) = \int_t^{t+\delta t} \frac{dF}{dX}(X(\tau)) dX(\tau) + \frac{1}{2} \int_t^{t+\delta t} \frac{d^2 F}{dX^2}(X(\tau)) d\tau.$$

I can now extend this result over longer timescales, from zero up to t , over which F does vary substantially to get

$$F(X(t)) = F(X(0)) + \int_0^t \frac{dF}{dX}(X(\tau)) dX(\tau) + \frac{1}{2} \int_0^t \frac{d^2 F}{dX^2}(X(\tau)) d\tau.$$



This is the integral version of **Itô's lemma**, which is usually written as

$$dF = \frac{dF}{dX} dX + \frac{1}{2} \frac{d^2 F}{dX^2} dt. \quad (5.5)$$

We can now answer the question, if $F = X^2$ what stochastic differential equation does F satisfy? In this example

$$\frac{dF}{dX} = 2X \quad \text{and} \quad \frac{d^2 F}{dX^2} = 2.$$

Therefore Itô's lemma tells us that

$$dF = 2X dX + dt.$$

This is *not* what we would get if X were a deterministic variable. In integrated form

$$X^2 = F(X) = F(0) + \int_0^t 2X dX + \int_0^t 1 d\tau = \int_0^t 2X dX + t.$$

Therefore

$$\int_0^t X dX = \frac{1}{2}X^2 - \frac{1}{2}t.$$

5.11 INTERPRETATION OF ITÔ'S LEMMA

Itô's lemma is going to be of great importance to us when we start to look at pricing options. If we can get comfortable with manipulating random quantities via simple rules of stochastic calculus then we will find most option theory quite straightforward.

To help in that regard, and to give you some insight into the role that Itô's lemma will be playing, take a look at the next figure, Figure 5.4.

In this figure you will see at the top a realization of a stock price, just a basic lognormal random walk. Below this is the value of an option on this stock.¹ What you will notice about these plots is that both have a direction to them (both are rising overall) and both have a random element (the bouncing around of the values).

Both look stochastic and we know that the stock price satisfies a stochastic differential equation

$$dS = \mu S dt + \sigma S dX,$$

so maybe the option value (call it $V(S, t)$) also satisfies a stochastic differential equation

$$dV = \underline{\hspace{1cm}} dt + \underline{\hspace{1cm}} dX.$$

The question is, then, what are the underlined bits? And that is precisely what Itô tells us.

This will be important later when we do the Black–Scholes theory, because knowing how much randomness there is in an option's value relative to a stock's value will give us a recipe for eliminating that randomness by buying an option and selling short a special quantity of the stock.

5.12 ITÔ AND TAYLOR

Having derived Itô's lemma, I am going to give some intuition behind the result and then slightly generalize it.

¹ It doesn't much matter whether it is a call, a put or something more exotic, the concept is relevant to all options.

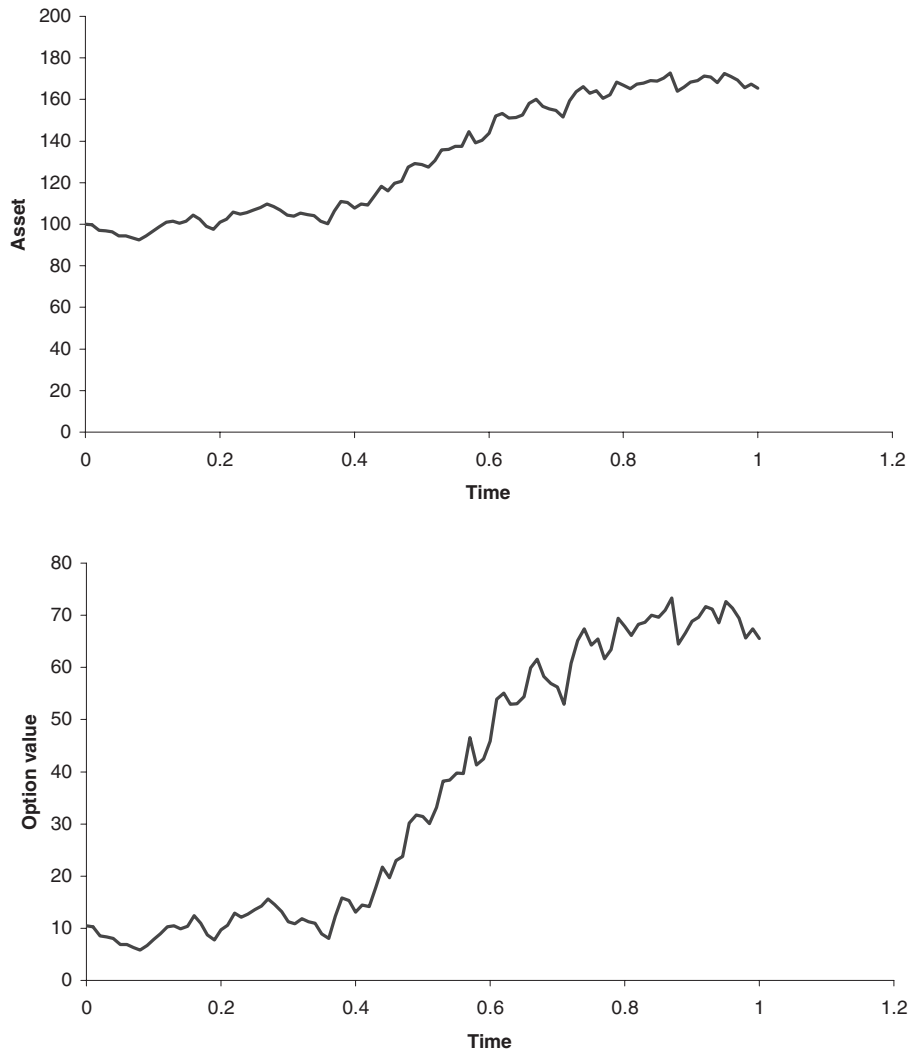


Figure 5.4 A realization of a stock price and the value of an option on that stock.

If we were to do a naive Taylor series expansion of F , completely disregarding the nature of X , and treating dX as a small increment in X , we would get

$$F(X + dX) = F(X) + \frac{dF}{dX}dX + \frac{1}{2}\frac{d^2F}{dX^2}dX^2,$$

ignoring higher-order terms. We could argue that $F(X + dX) - F(X)$ was just the 'change in' F and so

$$dF = \frac{dF}{dX}dX + \frac{1}{2}\frac{d^2F}{dX^2}dX^2.$$

This is very similar to (5.5) (and Taylor series *is* very similar to Itô), with the only difference being that there is a dX^2 instead of a dt . However, since in a sense

$$\int_0^t (dX)^2 = t$$

I could perhaps write

$$dX^2 = dt. \quad (5.6)$$

Although this lacks any rigor it does give the correct result. However, on a positive note you can, with little risk of error, use Taylor series with the ‘rule of thumb’ (5.6) and in practice you will get the right result. Although this is technically incorrect, you almost certainly won’t get the wrong result. I will use this rule of thumb almost every time I want to differentiate a function of a random variable.

Time Out...

Intuition behind $dX^2 = 'dt$

This is subtle. Pay close attention.

We shouldn’t really think of dX^2 as being the square of a single Normally distributed random variable, mean zero, variance dt . No, we should think of it as the sum of squares of lots and lots (an infinite number) of independent and identically distributed Normal variables, each one having mean zero and a very, very small (infinitesimal) variance. What happens when you add together lots of i.i.d. variables? In this case we get a quantity with a mean of dt and a variance which goes rapidly to zero as the ‘lots’ approach ‘infinity.’



To end this section I will generalize slightly. Suppose my stochastic differential equation is

$$dS = a(S) dt + b(S) dX, \quad (5.7)$$

say, for some functions $a(S)$ and $b(S)$. Here dX is the usual Brownian increment. Now if I have a function of S , $V(S)$, what stochastic differential equation does it satisfy? The answer is

$$dV = \frac{dV}{dS} dS + \frac{1}{2} b^2 \frac{d^2 V}{dS^2} dt.$$

We could derive this properly or just cheat by using Taylor series with $dX^2 = dt$. I could, if I wanted, substitute for dS from (5.7) to get an equation for dV in terms of the pure Brownian motion X :

$$dV = \left(a(S) \frac{dV}{dS} + \frac{1}{2} b(S)^2 \frac{d^2 V}{dS^2} \right) dt + b(S) \frac{dV}{dS} dX.$$

5.13 ITÔ IN HIGHER DIMENSIONS

In financial problems we often have functions of one stochastic variable S and a deterministic variable t , time: $V(S, t)$. If

$$dS = a(S, t) dt + b(S, t) dX,$$

then the increment dV is given by

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} b^2 \frac{\partial^2 V}{\partial S^2} dt. \quad (5.8)$$

Again, this is shorthand notation for the correct integrated form. This result is obvious, as is the use of partial instead of ordinary derivatives.

Occasionally, we have a function of two, or more, random variables, and time as well: $V(S_1, S_2, t)$. An example would be the value of an option to buy the more valuable out of Nike and Reebok. I will write the behavior of S_1 and S_2 in the general form

$$dS_1 = a_1(S_1, S_2, t) dt + b_1(S_1, S_2, t) dX_1$$

and

$$dS_2 = a_2(S_1, S_2, t) dt + b_2(S_1, S_2, t) dX_2.$$

Note that I have *two* Brownian increments dX_1 and dX_2 . We can think of these as being Normally distributed with variance dt , but *they are correlated*. The correlation between these two random variables I will call ρ . This can also be a function of S_1 , S_2 and t but must satisfy

$$-1 \leq \rho \leq 1.$$

The ‘rules of thumb’ can readily be imagined:

$$dX_1^2 = dt, \quad dX_2^2 = dt \quad \text{and} \quad dX_1 dX_2 = \rho dt.$$

Itô’s lemma becomes

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S_1} dS_1 + \frac{\partial V}{\partial S_2} dS_2 + \frac{1}{2} b_1^2 \frac{\partial^2 V}{\partial S_1^2} dt + \rho b_1 b_2 \frac{\partial^2 V}{\partial S_1 \partial S_2} dt + \frac{1}{2} b_2^2 \frac{\partial^2 V}{\partial S_2^2} dt. \quad (5.9)$$

5.14 SOME PERTINENT EXAMPLES

In this section I am going to introduce a few common random walks and talk about their properties.

Remember that a stochastic differential equation model for variable S is something of the form

$$dS = \text{---} dt + \text{---} dX.$$

The bit in front of the dt is deterministic and the bit in front of the dX tells us how much randomness there is. Modeling is very much about choosing functions to go where the underlining is, it is about choosing the functional form for the deterministic part and the functional form for the amount of randomness. We will now look at some examples.

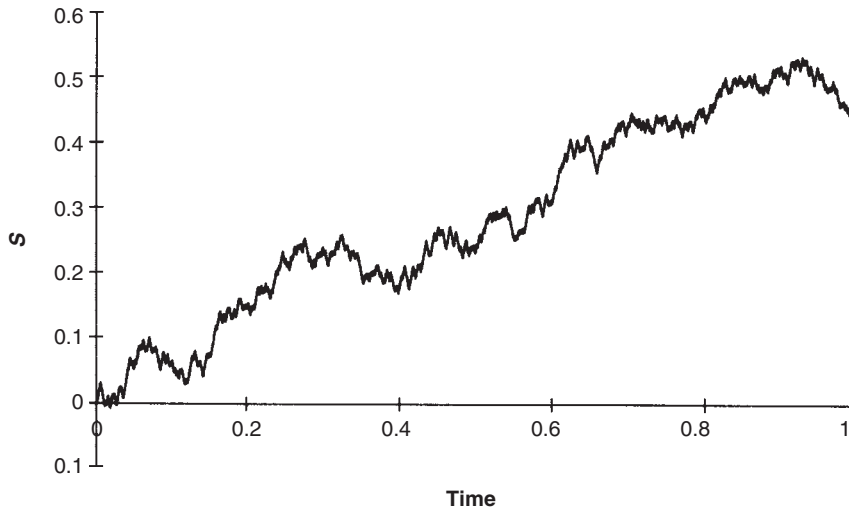


Figure 5.5 A realization of $dS = \mu dt + \sigma dX$.

5.14.1 Brownian motion with drift

The first example is like the simple Brownian motion but with a drift:

$$dS = \mu dt + \sigma dX.$$

A realization of this is shown in Figure 5.5. The point to note about this realization is that S has gone negative. This random walk would therefore not be a good model for many financial quantities, such as interest rates or equity prices. This stochastic differential equation can be integrated exactly to get

$$S(t) = S(0) + \mu t + \sigma(X(t) - X(0)).$$

Time Out...

Spreadsheet simulation

This random walk is simulated on the following spreadsheet.



	A	B	C	D	E	F
1	Start	100		Time	S	
2	μ	0.1		0	100	
3	σ	0.2		0.01	100.0171	
4			=B1	0.02	99.99066	
5	Timestep	0.01		0.03	99.96521	
6				0.04	99.97567	
7				0.05	99.93592	
8				0.06	99.95225	
9		=D8+\$B\$5		0.07	99.96202	
10				0.08	99.95202	
11				0.09	99.95474	
12				0.1	99.96519	
13		=E11+\$B\$2*\$B\$5+\$B\$3*SQRT(\$B\$5)*		0.11	99.94123	
14		(RAND()+RAND()+RAND()+RAND()+R		0.12	99.94159	
15		AND()+RAND()+RAND()+RAND()+RA		0.13	99.94805	
16		ND()+RAND()+RAND()+RAND()-6)		0.14	99.95993	
17				0.15	99.95776	
18				0.16	99.95448	
19				0.17	99.93744	
20				0.18	99.90828	
21				0.19	99.9203	
22				0.2	99.94772	



The spreadsheet is easily changed to model other random walks

5.14.2 The lognormal random walk

My second example is similar to the above but the drift and randomness scale with S :

$$dS = \mu S dt + \sigma S dX. \quad (5.10)$$

A realization of this is shown in Figure 5.6.

How does the time series in Figure 5.6 which was generated on a spreadsheet using random returns compare qualitatively with the time series in Figure 5.7 which is the real series for Glaxo-Wellcome?

If S starts out positive it can never go negative; the closer that S gets to zero the smaller the increments dS . For this reason I have had to start the simulation with a non-zero value for S . This property of this random walk is clearly seen if we examine the function $F(S) = \log S$ using Itô's lemma. From Itô we have

$$\begin{aligned} dF &= \frac{dF}{dS} dS + \frac{1}{2} \sigma^2 S^2 \frac{d^2 F}{dS^2} dt = \frac{1}{S} (\mu S dt + \sigma S dX) - \frac{1}{2} \sigma^2 dt \\ &= \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dX. \end{aligned}$$

This shows us that $\log S$ can range between minus and plus infinity but cannot reach these limits in a finite time, therefore S cannot reach zero or infinity in a finite time.

The integral form of this stochastic differential equation follows simply from the stochastic differential equation for $\log S$:

$$S(t) = S(0) e^{\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma (X(t) - X(0))}.$$

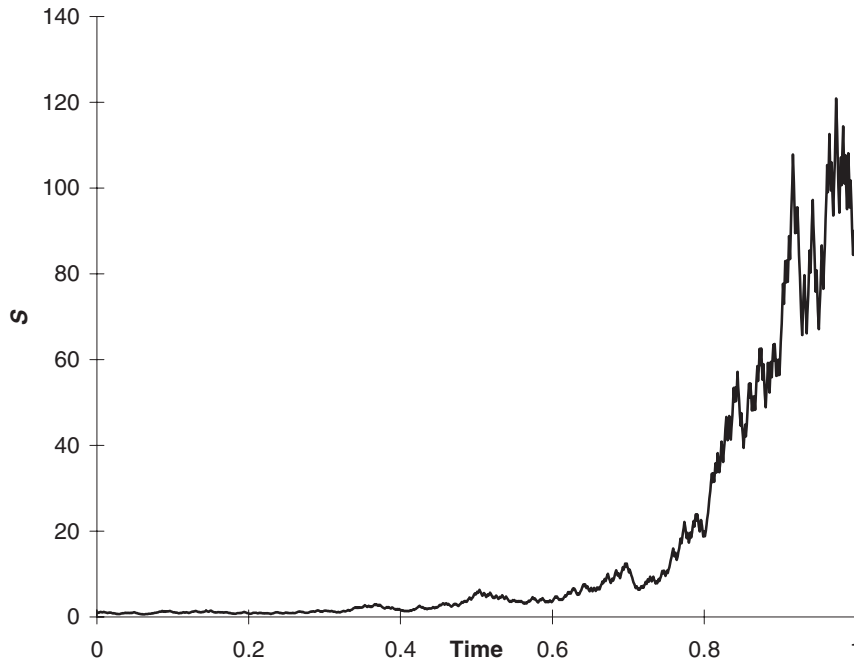


Figure 5.6 A realization of $dS = \mu S dt + \sigma S dX$.

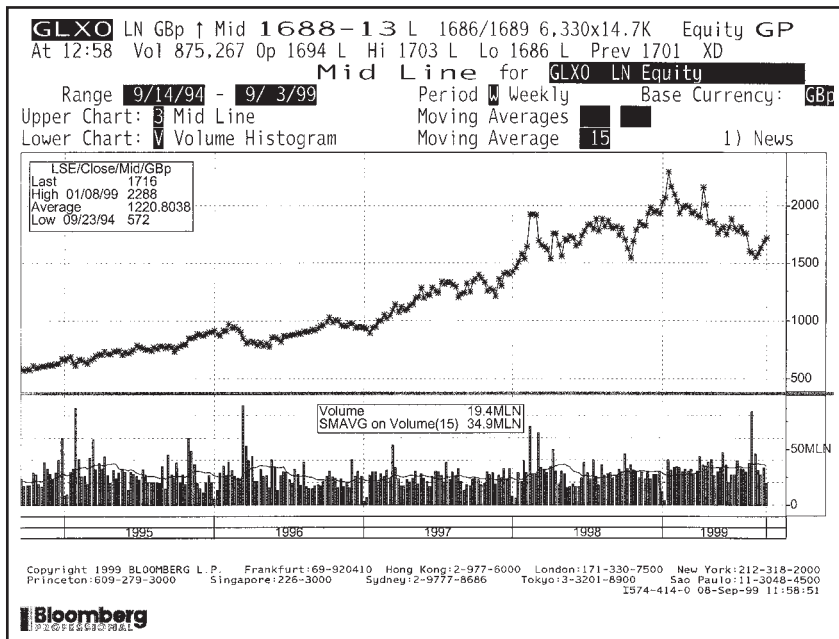


Figure 5.7 Glaxo-Wellcome share price (volume below). Source: Bloomberg L.P.

The stochastic differential equation (5.10) will be particularly important in the modeling of many asset classes. And if we have some function $V(S, t)$ then from Itô it follows that

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} dt. \quad (5.11)$$

5.14.3 A mean-reverting random walk

The third example is

$$dS = (\nu - \mu S) dt + \sigma dX.$$

A realization of this is shown in Figure 5.8.

This random walk is an example of a **mean-reverting** random walk. If S is large, the negative coefficient in front of dt means that S will move down on average, if S is small it rises on average. There is still no incentive for S to stay positive in this random walk. With r instead of S this random walk is the Vasicek model for the short-term interest rate.

Mean-reverting models are used for modeling a random variable that ‘isn’t going anywhere.’ That’s why they are often used for interest rates; Figure 5.9 shows the yield on a Japanese Government Bond.

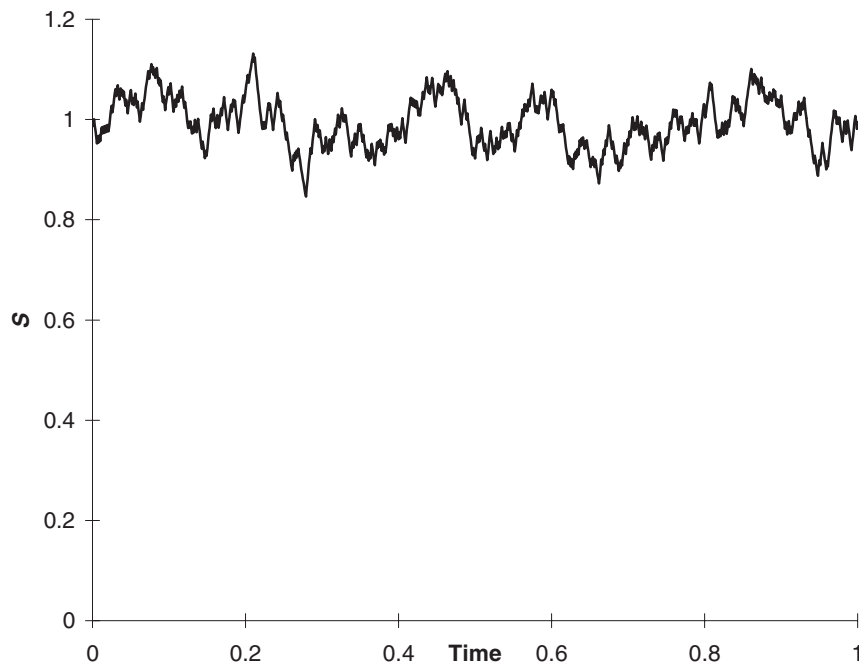


Figure 5.8 A realization of $dS = (\nu - \mu S) dt + \sigma dX$.

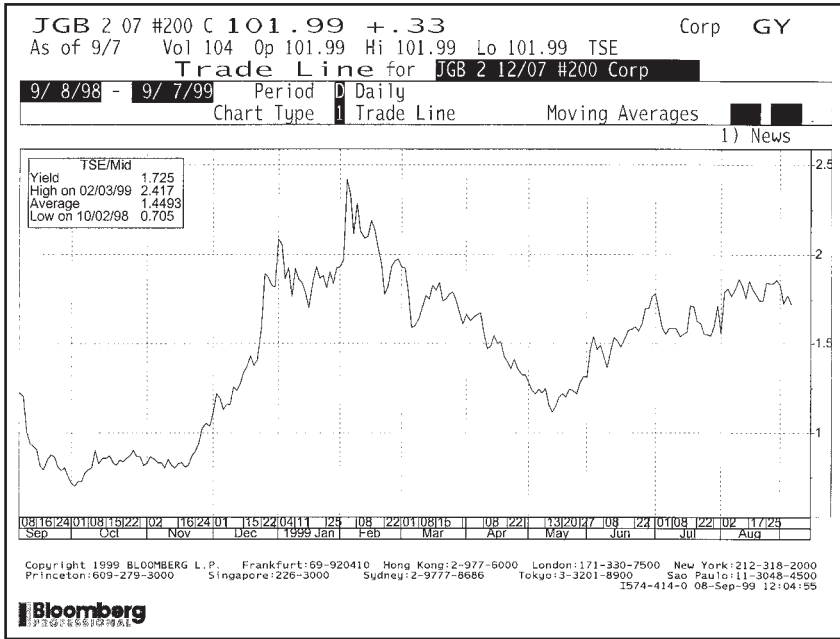


Figure 5.9 Time series of the yield on a JGB. Source: Bloomberg L.P.

5.14.4 And another mean-reverting random walk

The final example is similar to the third but I am going to adjust the random term slightly:

$$dS = (v - \mu S) dt + \sigma S^{1/2} dX.$$

Now if S ever gets close to zero the randomness decreases, perhaps this will stop S from going negative? Let's play around with this example for a while. And we'll see $\hat{I}t\hat{o}$ in practice.

Write $F = S^{1/2}$. What stochastic differential equation does F satisfy? Since

$$\frac{dF}{dS} = \frac{1}{2}S^{-1/2} \quad \text{and} \quad \frac{d^2F}{dS^2} = -\frac{1}{4}S^{-3/2}$$

we have

$$dF = \left(\frac{4v - \sigma^2}{8F} - \frac{1}{2}\mu F \right) dt + \frac{1}{2}\sigma dX.$$

I have just turned the original stochastic differential equation with a variable coefficient in front of the random term into a stochastic differential equation with a constant random term. In so doing I have made the drift term nastier. In particular, the drift is now singular at $F = S = 0$. Something special is happening at $S = 0$.

Instead of examining $F(S) = S^{1/2}$, can I find a function $F(S)$ such that its stochastic differential equation has a zero drift term? For this I will need

$$(v - \mu S) \frac{dF}{dS} + \frac{1}{2}\sigma^2 S \frac{d^2F}{dS^2} = 0.$$

This is easily integrated once to give

$$\frac{dF}{dS} = AS^{-\frac{2\nu}{\sigma^2}} e^{\frac{2\mu S}{\sigma^2}} \quad (5.12)$$

for any constant A . I won't take this any further but just make one observation. If

$$\frac{2\nu}{\sigma^2} \geq 1$$

we cannot integrate (5.12) at $S = 0$. This makes the origin **non attainable**. In other words, if the parameter ν is sufficiently large it forces the random walk to stay away from zero.

This particular stochastic differential equation for S will be important later on, it is the Cox, Ingersoll & Ross model for the short-term interest rate.

These are just four of the many random walks we will be seeing.

5.15 SUMMARY

This chapter introduced the most important tool of the trade, Itô's lemma. Itô's lemma allows us to manipulate functions of a random variable. If we think of S as the value of an asset for which we have a stochastic differential equation, a 'model,' then we can handle functions of the asset, and ultimately value contracts such as options.

If we use Itô as a tool we do not need to know why or how it works, only how to use it. Essentially all we require to successfully use the lemma is a rule of thumb, as explained in the text. Unless we are using Itô in highly unusual situations, then we are unlikely to make any errors.

FURTHER READING

- Neftci (1996) is the only readable book on stochastic calculus for beginners. It does not assume any knowledge about anything. It takes the reader very slowly through the basics as applied to finance.
- Once you have got beyond the basics, move on to Øksendal (1992) and Schuss (1980).



Time Out...

Stochastic calculus for dummies . . . learning by using

To use stochastic calculus successfully only really requires a little bit of intuition. Then, with use, familiarity breeds, if not contempt, sufficient confidence to make you believe that you understand. (And that's how I've learned the subject.) So, here is that intuition.

- Stochastic differential equations are like recipes for generating random walks, just as we saw in the previous chapter using Excel.
- If you have some quantity, let's call it S , that follows such a random walk, then any function of S is also going to follow a random walk. For example, if S is moving about randomly, then so is S^2 .
- The question then becomes 'What is the random walk for this function of S ?' That is, what is its recipe, or what is its stochastic differential equation?
- The answer to that comes from applying something very like Taylor series but with two tricks.
- The first trick is that when you do your Taylor series expansion, only keep terms of size dt or bigger ($dt^{1/2}$).
- The second trick is that every time you see a dX^2 term replace it with dt . Why? Because dX is really made up of lots of Normally distributed random variables, and so dX^2 becomes its expected value dt .

EXERCISES

In all of these $X(t)$ is Brownian motion.

1. By considering $X^2(t)$, show that

$$\int_0^t X(\tau) dX(\tau) = \frac{1}{2}X^2(t) - \frac{1}{2}t.$$

2. Show that

$$\int_0^t \tau dX(\tau) = tX(t) - \int_0^t X(\tau) d\tau.$$

3. Show that

$$\int_0^t X^2(\tau) dX(\tau) = \frac{1}{3}X^3(t) - \int_0^t X(\tau) d\tau.$$

4. Consider a function $f(t)$ which is continuous and bounded on $[0, t]$. Prove integration by parts, i.e.

$$\int_0^t f(\tau) dX(\tau) = f(t)X(t) - \int_0^t X(\tau) df(\tau).$$

5. Find $u(W, t)$ and $v(W, t)$ where

$$dW(t) = u dt + v dX(t)$$

and

(a) $W(t) = X^2(t),$

- (b) $W(t) = 1 + t + e^{X(t)}$,
 (c) $W(t) = f(t)X(t)$,

where f is a bounded, continuous function.

6. If S follows a lognormal random walk, use Itô's lemma to find the differential equations satisfied by

- (a) $f(S) = AS + B$,
 (b) $g(S) = S^n$,
 (c) $h(S, t) = S^n e^{mt}$,

where A, B, m and n are constants.

7. If $dS = \mu S dt + \sigma S dX$, use Itô's lemma to find the stochastic differential equation satisfied by $f(S) = \log(S)$.

8. The change in a share price satisfies

$$dS = A(S, t) dX + B(S, t) dt,$$

for some functions A, B , what is the stochastic differential equation satisfied by $f(S, t)$? Can A, B be chosen so that a function $g(S)$ has a zero drift, but non-zero variance?

9. Two shares follow geometric Brownian motions, i.e.

$$dS_1 = \mu_1 S_1 dt + \sigma_1 S_1 dX_1,$$

$$dS_2 = \mu_2 S_2 dt + \sigma_2 S_2 dX_2.$$

The share price changes are correlated with correlation coefficient ρ . Find the stochastic differential equation satisfied by a function $f(S_1, S_2)$.